Process Improvement Opportunities Evaluation

While Task 1 focused on a qualitative definition of the process, this task will use data-driven insights derived from Celonis, an enterprise process mining platform. Process mining uses event log data to reconstruct a process, measure performance, and identify deviations from the expected process. This approach revealed differences between initial stakeholder perceptions and operational reality.

A. Business Process Summary

Udon-ya initially identified 9 process steps shared across three stakeholder groups: management, employees, and customers. However, analysis shows that there are 36 distinct steps in the process.

Additionally, process mining revealed that there is a separate process path for the kitchen staff, as opposed to the servers up front. The employee stakeholder group actually consists of two staff groups: the kitchen and service staff.

The table below shows the account of activities mapped to each of the 9 original steps:

Process Step	Count of Activities
1. Restaurant preps food	**11**
2. Customers enter the restaurant	1
3. Customers are seated	1
4. Restaurant receives food order	1
5. Restaurant cooks food	**11**
6. Food is served to customers	1
7. Customers eat	1
8. Customers pay	1
9. Customers leave the restaurant	1
Total:	29

There are also 7 additional activities related to store opening and closing that are not captured in the as-is process, totaling 36 distinct activities.

Challenges and inefficiencies

Udon-ya is currently experiencing four main business process challenges and inefficiencies:

1. **High customer volume**: The Udon-ya team struggles when customer volume is high.
2. **Takeout orders:** A lack of prioritization in pick-up order fulfillment is causing longer wait times for customers.
3. **Insufficient food preparation**: Inconsistencies surrounding the availability of prepped foods for order fulfillment frustrate the Udon-ya team, leading to increased customer wait times.
4. **Customer feedback**: Udon-ya currently lacks a method to track customer satisfaction, and therefore cannot effectively address customer concerns.

Performance Metrics Summary

Udon-ya has previously identified the following metrics of interest. Process mining and analysis using Celonis yielded actual current performance metrics, including throughput time. Throughput is the time it takes to complete one, a series, or a cycle of activities.

1. Average Time from Order to Table

This metric reflects the time it takes to deliver food to the customer from the moment they place the order, directly impacting customer wait times, with a target of under 7 minutes. Since this metric examines the order process as a whole, increases (above 7 minutes) in this metric can indicate gaps or inefficiencies in the overall order process.

Below is a histogram showing the distribution of average throughput time from receiving the order to serving the customer. The current average throughput time is 7.6 minutes, which is 0.7 minutes over target.

2. Food Preparation (Cooking) Time

This metric reflects the total time spent cooking an order from the moment the restaurant receives an order to when the order is finished cooking, with a target of under 5 minutes. Food cooking time identifies issues that occur specifically in the cooking part of the process.

Average cooking throughput time is 6.38 minutes, or 1.37 minutes over target:

3. Average Time from Preparation to Delivery

This metric represents the time elapsed from the end of cooking the food to delivery to the customer. Ready-to-deliver orders should not sit for longer than 2 minutes before being delivered to the customer. This metric identifies process issues that occur just before delivery to the customer.

Udon-ya's average throughput time from garnishing to order delivery is 2.44 minutes, which is 0.45 minutes over target.

Business process framework

Lean Framework

According to Lean Enterprise Institute (2025), Lean is a process improvement methodology that focuses on delivering value to customers while eliminating waste through continuous process improvement. Lean methodology addresses questions about purpose, process, and people to seek solutions to delivery problems.

In Lean processes, waste refers to process inefficiencies and can be considered any of the following:

1. Defects – errors requiring correction
2. Overproduction – Making more product than what is needed
3. Waiting – Idle time during the process
4. Non-utilized or underutilized talent – not using human resources effectively
5. Transportation – Materials are not co-located and need relocation
6. Inventory – excess materials or not enough materials
7. Motion – Unnecessary movement of human resources
8. Excessive processing – Doing more work than is necessary for the task

These eight types of waste provide a standardized framework for identifying process inefficiencies. The goal of a Lean process is perfection, meaning the process, process participants, and especially customers receive what they need, when they need it, with no wasted effort or resources.

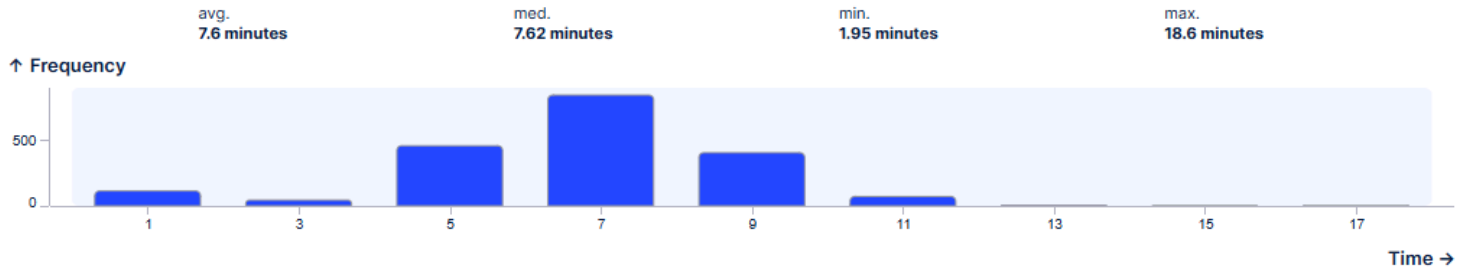
B. Two underperforming key performance metrics

Based on the gathered throughput times and significance of the metrics, I decided to target two underperforming metrics: Average Time from Order to Table and Food Preparation (Cooking) Time.

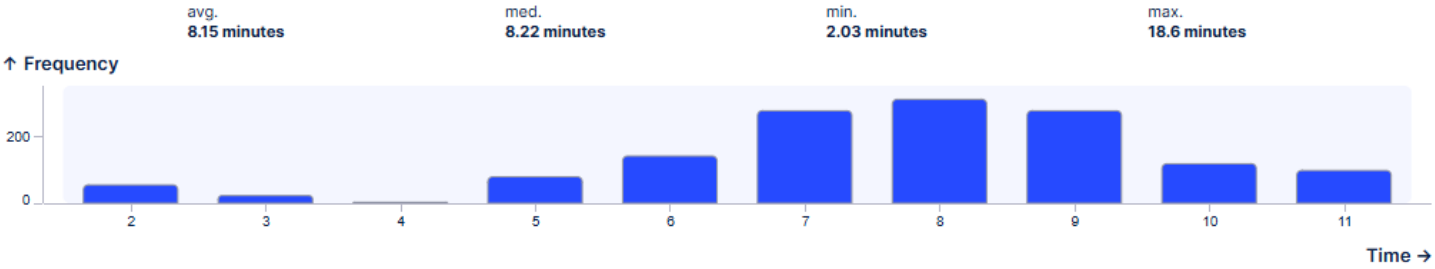
1) Average Time from Order to Table is a foundational metric that monitors the overall health of the process and acts as an early indication of process breakdowns.

The following are three visualizations representing throughput time metrics for normal, non-peak, and peak orders.

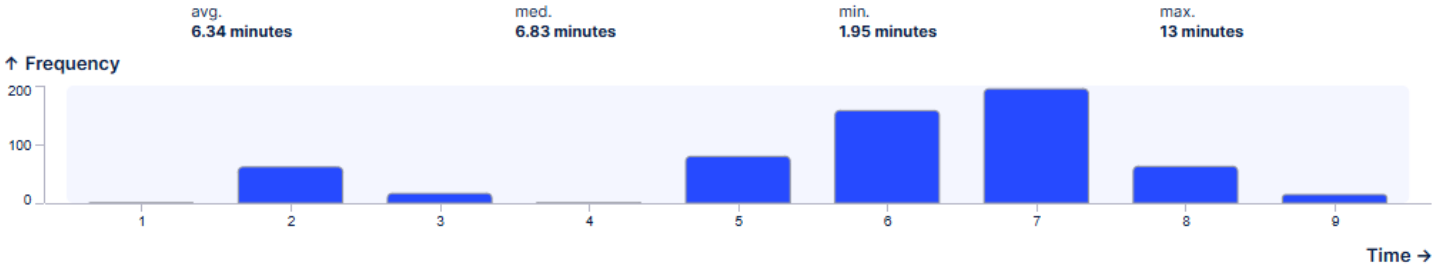
Normal - Average Time from Order to Table



Non-Peak - Average Time from Order to Table



Peak - Average time from Order to Table



As can be seen from the analysis, the restaurant is under target for peak orders at 6.34 minutes, while non-peak orders are over target at 8.15 minutes. These throughput times suggest that the root cause

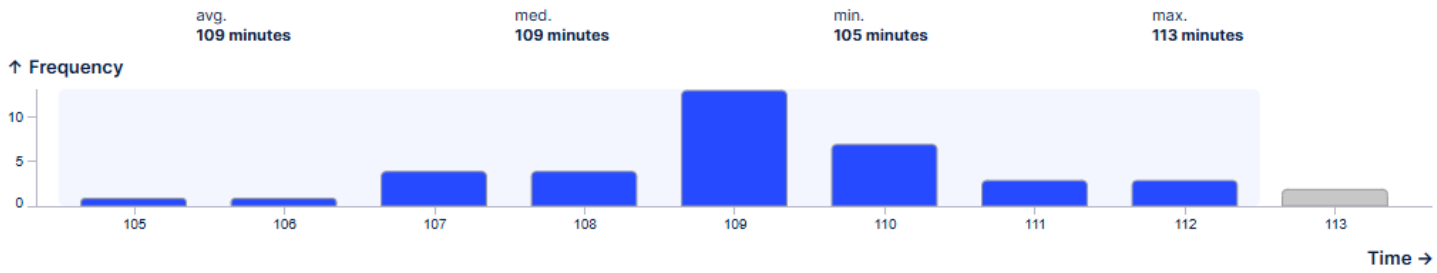
of perceived inefficiencies during peak customer volume periods is not related to order volume. Instead, the issue lies in the process itself, since this metric performs worse during normal operations.

Additionally, the maximum throughput time values of 18.6 minutes (non-peak) and 13 minutes (peak) represent instances where the order process broke down. The variance between target and maximum throughput times indicates variability in process execution.

2) Food Preparation (Cooking) Time measures the efficiency and coordination of activities between kitchen employees for ingredients and service staff for cooking.

Currently, cooking time only accounts for activities in the service employee path, which is presently 6.38 minutes and exceeds the target. Kitchen ingredients preparation, which occurs separately from the order process, has an average throughput time of 109 minutes.

Avg kitchen ingredients preparation TPT



This analysis highlights both the lack of integration between kitchen and server activities as well as potential process issues isolated to kitchen ingredient preparation, which lead to delayed or insufficient ingredient preparation.

C. Business Process Methodology

Addressing identified inefficiencies with Lean

The Lean methodology directly addresses Udon-ya's identified inefficiencies by focusing on eliminating waste and coordinating processes. The process mining analysis revealed specific inefficiencies that meet the definition of waste under Lean, such as variability in the overall order process and a lack of integration between the kitchen and servers.

Udon-ya's ultimate goal is to design a standardized process for franchising. Lean addresses inconsistent process execution through systemic process optimization and waste elimination.

- **Lean tools to improve the process**

Lean offers several concepts and tools for achieving the goal of a standardized process without waste. The following is a discussion of Lean techniques that apply to Udon-ya's process, which provide paths for eliminating process inefficiencies (Lean Enterprise Institute, 2025).

<u>The Eight Wastes</u>
As discussed in section A, Lean provides a guide for classifying types of waste. Based on the identified inefficiencies of variability and lack of process coordination, Udon-ya is experiencing a few types of waste in the order process. Lean provides solutions based on the type of wastes identified.

Waiting
Any time in the process when service staff or kitchen staff are not actively engaged in an activity, they are waiting. Waiting represents unproductive time during the process and is a target for elimination or redirection. This type of waste is evident in the service staff's path before customer arrival and while customers are eating. Kitchen staff prepare a single batch of ingredients and then are idle for the remainder of the process.

Inventory
Since ingredient preparation activities occur only once a day, the restaurant is at the mercy of customer volume at any given time, as it sets an arbitrary limit on the availability of prepared ingredients. This ingredient scarcity issue creates two forms of inventory waste: excess ingredients represent waste when there are fewer customers, and insufficient ingredients result in waste when customer demand is high.

Motion
Currently, the kitchen staff and service staff are separated both in terms of process and physically. Kitchen staff have no idea what service staff is doing, and vice versa. Lack of staff coordination results in waste by requiring repeated non-value-added tasks, which represent unnecessary effort. An integrated process prevents motion waste.

Non-utilized or underutilized talent
Customers generate revenue for the restaurant, but there is currently no visibility into the customer experience. Talent waste occurs when the order process fails to benefit from customer-driven improvements.

Continuous Flow
Lean methodology emphasizes the concept of continuous flow, where each product is incrementally assembled within an integrated workflow from beginning to end (Lean Enterprise Institute, 2025). Each step of the process contributes to or generates one part of the product in sequential order. Adapting a continuous flow process will help Udon-ya identify missing or unnecessary steps and duplicative efforts leading to waste.

Pull Systems
According to Lean Enterprise Institute (2025), pull systems are beneficial for inventory management, ensuring that customer demand aligns with the availability of raw materials needed for product assembly. Pull systems typically utilize a signaling system to trigger the need for additional materials. A pull system would enable Udon-ya to establish a connection between the kitchen and service staff, ensuring that prepared ingredients are always available while minimizing excess inventory waste.

Kaizen
A core Lean principle is continuous improvement, also known as Kaizen, which aims to proactively eliminate waste through constant monitoring and examination of process inefficiencies observed by any stakeholder in the process. Kaizen requires a standardized process to serve as the baseline against which to measure improvements. Once the order process is standardized, Udon-ya will need to continuously monitor process performance, especially during and after franchise adoption.

Lean for underperforming metrics

The application of these Lean tools will improve Udon-ya's underperforming metrics by addressing their root causes. Currently, Food Preparation Time exceeds the target of under 5 minutes due to

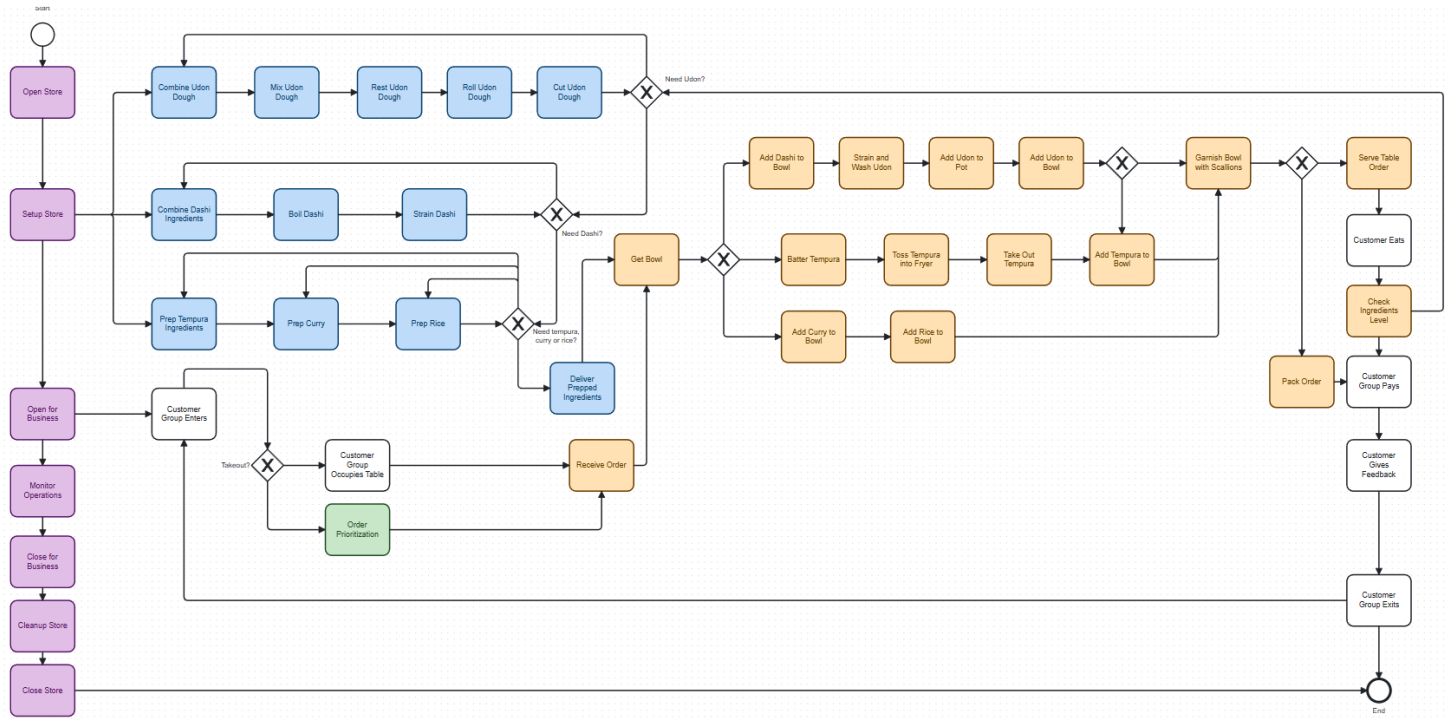
coordination issues between the kitchen and service staff. Continuous flow and a pull system will eliminate these delays by ensuring that the right amount of ingredients is prepped and available when needed.

Average Time from Order to Table is also above the target of under 7 minutes. Implementing standardized work to create a standardized and wasteless process that will improve this metric. Standardized work will reduce the maximum throughput time, which will inherently reduce the average throughput time.

D. Opportunities for Process Improvement

The process map below (also provided separately with this submission, named NewUdonProcess.pdf) shows the proposed continuous flow order process for Udon-ya. Purple rectangles represent management activities, blue for kitchen activities, orange for service activities, white for customer activities, and green for activities not explicitly assigned to a stakeholder.

Proposed improved process model



Specific Process Improvements

Lean waste elimination

Waiting - The improved process model implements a continuous cycle for ingredient preparation. This new workflow ensures that the kitchen continues to prepare ingredients until no ingredients are needed. In the model, as demonstrated by the cycle loops employed at the ends of the blue workflows, which connect back to the beginning of each preparation workflow. The continuous cycle eliminates waiting waste for all stakeholders.

Inventory – The improved process model introduces a specific service staff action (Check Ingredient Levels) to create an ingredients pull system, which creates a feedback loop between the kitchen and

service staff. With this pull system, the kitchen staff will be able to match ingredient inventory to customer demand, eliminating inventory waste.

Motion – The improved process model integrates kitchen activities and service activities into a single process, enabling communication efficiencies between the kitchen and service teams. In the model, all kitchen prep activities flow into the "Get Bowl" activity, and the "Check Ingredient Levels" activity feeds back into the ingredients preparation cycle. This coordination eliminates the waste of inconsistent communication and the associated non-value-added activities surrounding ingredient replenishment.

Non-utilized talent – The improved process model introduces a specific customer action (Customer Gives Feedback) to enable consistent gathering of customer feedback. Gathering this feedback will provide Udon-ya with additional insight into the process's performance from the customer's perspective, eliminating the waste of untapped customer knowledge that Udon-ya can utilize in future Kaizen process improvement endeavors.

Challenges and issues

The improved process model directly addresses Udon-ya's current challenges and issues:

High customer volume: The Udon-ya team struggles when customer volume is high.

The process mining analysis revealed that delays were not occurring for peak-time customers, and as a result, the delays appeared to be related to process inefficiencies. The continuous flow, combined with pull systems, addresses this challenge by eliminating bottlenecks that add to order delays during busy periods. The integrated workflow process maintains ingredient preparation cycles, while the pull system provides real-time inventory management. Finally, the new communication loops will help alleviate process breakdowns, which employees likely perceive as more chaotic when demand increases.

Takeout orders: A lack of prioritization in pick-up order fulfillment is causing longer wait times for customers.

As part of the continuous flow design, the improved model includes a classification gateway immediately after the Customer Group Enters activity. This gateway checks whether or not the order is a takeout order. If not, then customers are seated. If so, the customer bypasses the seating path and proceeds directly to the service path. This gateway eliminates waiting waste for both takeout and dine-in customers. The separate routing ensures prioritization of takeout orders without affecting dine-in service.

Insufficient food preparation: Inconsistencies surrounding the availability of prepped foods for order fulfillment frustrate the Udon-ya team, leading to increased customer wait times.

The improved continuous flow process and ingredients pull system directly addresses insufficient food preparation by replacing the single daily ingredients preparation workflow with a both proactive and responsive workflow that revolves around ingredient availability and real-time customer demand. Through this improved process, the service staff are in constant communication with kitchen staff, eliminating inconsistencies in ingredient availability.

Customer feedback: Udon-ya currently lacks a method to track customer satisfaction, and therefore cannot effectively address customer concerns.

The addition of the Customer Gives Feedback activity addresses this challenge by creating a systematic way to collect feedback. Consistent collection of feedback will enable Udon-ya to create a continuous improvement loop with customers, gaining additional insight into how the process is performing. This approach ensures that Udon-ya can gather data, monitor metrics, and implement customer-driven improvements along the way via continuous improvement.

Impacts on performance metrics and business performance

Performance Metrics Impact

Average Time from Order to Table: The standardized workflow and communication loops will reduce process variability, addressing and eliminating the divergence between peak (6.34 minutes) and non-peak (8.34 minutes) order delays. The kitchen-service coordination will address order-to-table time variations, which indicate process breakdowns, and ensure consistent performance across all levels of customer demand. Combined with other process optimizations, these improvements will reduce Average Time from Order to Table.

Food Preparation Time: The continuous flow and pull system improvements will reduce Food Preparation Time by eliminating coordination delays between the kitchen and service teams. The integrated workflow ensures that ingredients are available before receiving orders, rather than reacting to them by preparing ingredients reactively. This effort prevents the delays resulting from not continuously restocking ingredients throughout the day.

Overall Business Performance

The improved process aligns with Lean Enterprise Institute's (2025) Lean principles for creating a standardized, repeatable, and scalable process required for franchising. The benefits of implementing this lean process for Udon-ya include 1) standardized workflows that eliminate process variability, 2) systematic process waste elimination, and 3) continuous feedback loops with the customer. These combined improvements set the foundation for franchising success by applying the same optimized process to future locations.

REFERENCES

Lean Enterprise Institute. (2025). Lexicon Terms. *Lean.org.* https://www.lean.org/explore-lean/lexicon-terms/